

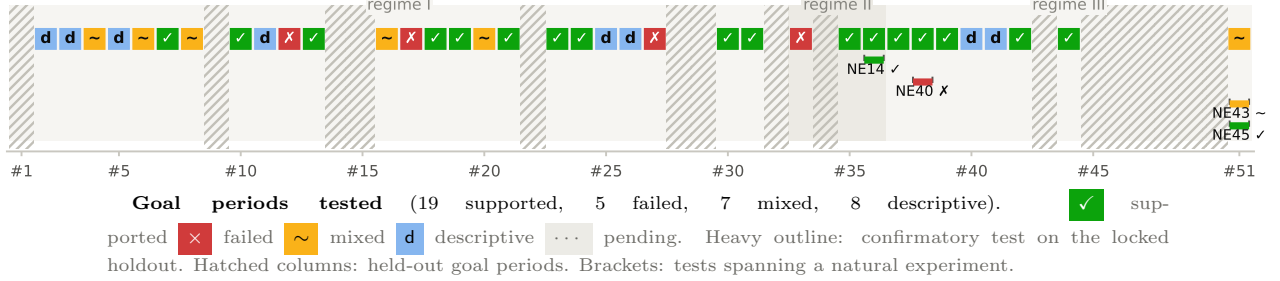
H74: An operator-grade change detector

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 45% Faithfulness 2/5 Usefulness 3/5

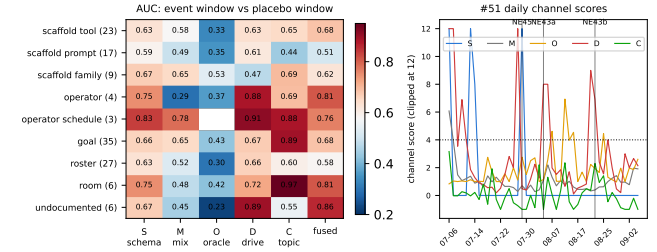
Bottom line. A schema diff plus operator counters is a precise, nearly free platform-change monitor; the fused five-channel alarm is too noisy as tuned.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

A daily change-point monitor runs on all 282 non-holdout active days, from April 2025 to September 2026. It has five channels: log record schemas (S), synchronous within-agent behavior shifts (M), search-answer format (O), drive and schedule counters (D) and content topic shift (C). The fused alarm is the maximum channel score at $\tau = 4$. It is scored against 211 catalogued changes and 6 undocumented ones. On documented tool changes the fused score has AUC 0.68 [0.54, 0.82] (random-date p 0.059): **mixed**. It alarms on 18% of placebo days, against a predicted 10%. The schema channel has no false alarms and dates platform changes to the day, naming the changed fields. It also finds six provider API format changes that no catalog lists. Drive counters date both NE43 steps. Topic shift remains the goal detector. The holdout is not run.



Observables. Left: AUC of the event window (days $-1..+1$) against placebo windows, by event class and channel; event counts in brackets. Each channel sees its own class: schema and drive for platform and operator steps, topic for goals and rooms. The fused score averages them away. Right: #51 daily channel scores. The schema channel fires on the search-tool rename (NE45). The drive channel fires on the bookend stop (NE43a) and the nudger stop (NE43b).

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

PHYSICAL MODEL

Models 02 and 11. A day d is one step of a driven system. A platform step changes the rules for every agent at once, so the record schema changes or every agent's own behavior shifts together. An operator step changes the drive counters. A goal step is a field step on content. Each feature is scored by a robust trailing z against the previous $B = 10$ days. c_k is the Gaussian consistency factor of the trimmed SD. n_d^{new} and n_d^{retired} count platform-wide record signatures (field names and value types) that appear or retire on day d .

$$z(d) = [x(d) - \text{med}_B x] / \max\{\text{sd}_B^{\text{trim}}(x) / c_k, \text{floor}\}$$

$$z_M(d) = \max_f |\text{med}_i z_{i,f}(d)|, \quad z_S(d) = 4(n_d^{\text{new}} + n_d^{\text{retired}})$$

$$Z(d) = \max\{z_S, z_M, z_O, z_D, z_C\}, \quad \text{alarm if } Z(d) \geq \tau = 4$$

Run a schema diff on your logs. Hash the field names and value types of every record each day, and alarm when a signature used by ≥ 2 agents appears or retires. Here it gave 0 false alarms in 34 quiet days. It dated the regime change, the outreach tool and an undocumented search-tool rename to the day, and it found six silent provider API changes. Add plain counters for operator drives (bookends, nudges, hours, joint silence): they dated both NE43 steps. Use topic shift for goal changes (AUC 0.89). Do not fuse heavy-tailed behavior statistics with a Gaussian threshold: they fire on 6–9% of quiet days each. Prompt-only changes stay invisible to every channel.

RESULTS VS PREDICTIONS

Prediction (dated)	Observed	Verdict
P0 synthetic own-channel hit ≥ 0.8	S 1.00, D 0.97, C 0.98, O 0.80, held M 0.90 (5 SD; 0.05 at 3 SD); FAR 0.051	
P1 tool changes: AUC ≥ 0.70 , $p < 0.05$	AUC 0.68 [0.54, 0.82]; hit 0.78; mixed p 0.059	
P2 undocumented $\geq 4/6$	6/6 in window (p 0.13); 4/6 at-tributable	held, weak
P3 goals: C AUC ≥ 0.9 , $S \leq 0.6$	0.89; 0.66	failed (narrowly)
P4 fused FAR ≤ 0.10	0.18 per day (S 0, C 0)	failed
P5 fused beats topic shift	undocumented +0.30 [0.01, 0.59]; tools +0.04; goals -0.16	failed
P6 roster: hit ≥ 0.3 , $S \leq 0.2$	0.63; 0.26	half
NE14 regime change	S = 16 on 03-24 (new pause, supported search schemas)	
NE45 search-tool rename	S = 12 on 07-29; S at 1/8 #51 joins	supported
NE43 two operator steps	D = 8 (08-05), 6.9 (08-21); quiet days 2/7	mixed
NE40 oracle swap	O ≤ 2.8 : missed	failed

synthetic hit rate at tau=4 (fused FAR/day 0.051)

schema	S	M	O	D	C	fused
schema	1.00	0.00	0.05	0.05	0.02	1.00
mix5	0.00	0.90	0.03	0.17	0.00	0.93
mix3	0.00	0.05	0.05	0.15	0.02	0.25
oracle	0.00	0.00	0.80	0.07	0.02	0.80
drive	0.00	0.00	0.08	0.97	0.00	0.97
content	0.00	0.00	0.10	0.07	0.98	0.98
roster	0.00	0.00	0.08	0.22	0.03	0.32
goalmix	0.00	0.00	0.03	0.12	0.98	0.98

Second figure. Synthetic daily streams at real counts (20 runs, 24 planted steps each). Each channel catches its own step type at a fused per-day false-alarm rate of 0.05. The mix channel needs synchronous shifts of about 5 within-agent SD. The real false-alarm rate was 3.5 $\times$ the synthetic one: real noise has heavier tails.

GOAL PERIODS IN DETAIL

G02 d descriptive
 G03 d descriptive
 G04 $\sim$ mixed
 G05 d descriptive
 G06 $\sim$ mixed
 G07 $\checkmark$ supported
 G08 $\sim$ mixed
 G10 $\checkmark$ supported
 G11 d descriptive
 G12 $\times$ failed
 +29 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	2
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	1
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

Only 34 placebo days survive the dense catalog (FAR 6/34, Wilson 95% 0.08–0.33). NE40 and NE45 were found by hand by H56, so they are recoveries, not blind discoveries. With 30% of days in alarm, period-level “hits” are weak evidence. Provider API changes show in S, but they are not village platform changes. One baseline length and one threshold were tried. The evaluation catalog is H56’s CHANGELOG parse. The holdout is not run.

ROUND 2

Build a precision monitor from S, D counters and C, with per-channel thresholds at 2% FAR. Add a presence/retirement rule for format markers (NE40). Use empirical-quantile baselines.

- **Essence.** a cheap monitor that tells an operator, on the day, that the platform, the drive or the goal changed, with a known false-alarm rate.
- **H74-R1** Precision monitor.
- **H74-R2** Presence/retirement rule for formats.
- **H74-R3** Heavy-tailed baselines.
- **H74-R4** Provider API changes as natural experiments.
- **H74-R5** Holdout.